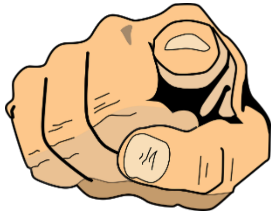
Semester Projekt: Blutzucker-Insulin Digital Twin

- In den letzten Jahren sind CGM-Patches (Continuous Glucose Monitoring) allgemein verfügbar geworden.

- Ein CGM-Patch misst den Glukosewert im Gewebe (Interstitium).
  Wir werden sehen, dass sich daraus der Blutzucker- und Insulinspiegel im Blut abschätzen lässt.

- Blutzucker und Insulin sind entscheidend für Wohlbefinden und Energiehaushalt des Menschen.

- Sie werden unter anderem von Diabetikern und SportlerInnen überwacht, um Gesundheit und Leistungsfähigkeit zu verbessern.

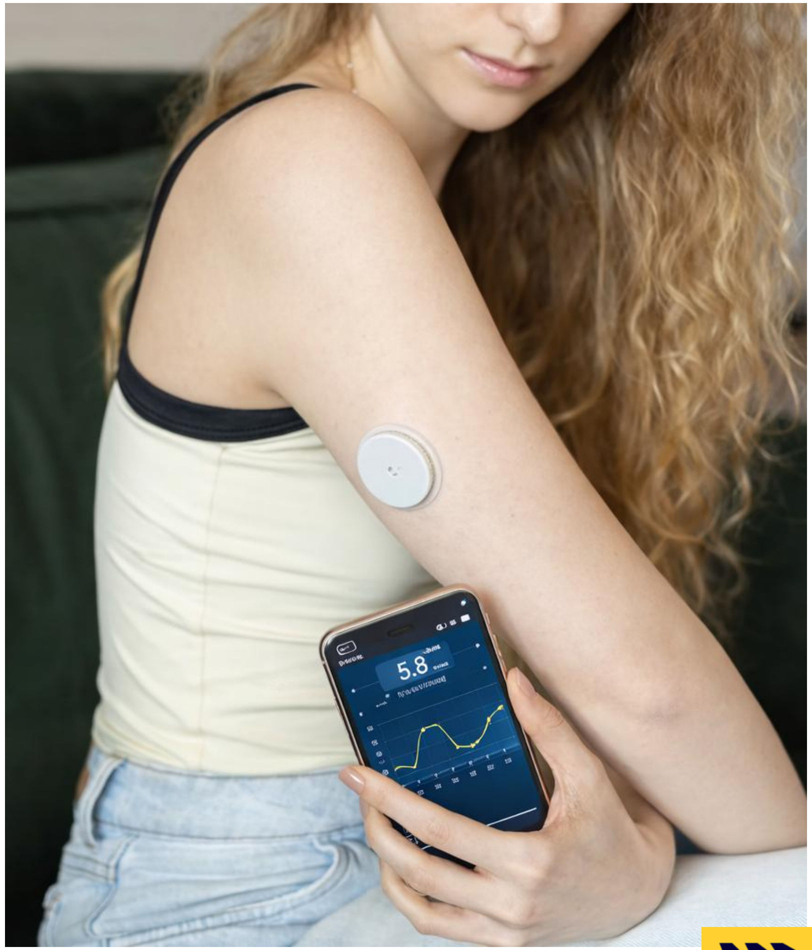


Semester Projekt: Blutzucker-Insulin Digital Twin

- Macht euch mit dem Thema vertraut, indem ihr eure **eigenen Erfahrungen und Erwartungen** reflektiert.

- Erstellt ein erstes Brainstorming zum **Glukose-Insulin-System**:
  - Was wisst ihr bereits?
  - Was ist euch unklar?
  - Wie könnte euch das Wissen über eure Glukose- und Insulinwerte helfen?

- Dokumentiert eure Antworten im **Miro-Board**.



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Semester Projekt: Blutzucker-Insulin Digital Twin

For the Implementation of the Digital Twin, you may use MATLAB 2025b or Python

For MATLAB:

1. Install MATLAB on your computer (Campus License)
   - Minimum: MATLAB & SIMULINK. Further toolboxes later, when required.

2. Check out the Mathworks academy: MATLAB Onramp
   There is also a SIMULINK Onramp. This includes elements from SW2

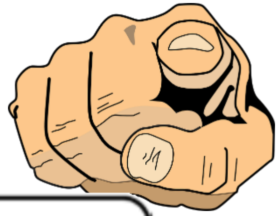
MathWorks®

For Python:

1. Choose your development environment – e.g. PyCharm orJupiter Notbooks

2. Familiarize yourself with Python libraries for simulating differential equations and graphing time series.

If you run into problems or have questions, please note these in miro, or bring them to the discussion in SW02.



Digital Twin Development

- This week we start the modeling process for the Glucose-Insulin System of a Person.

- The goal is to model the effect of a meal on the Glucose and Insulin levels in the Blood Plasma

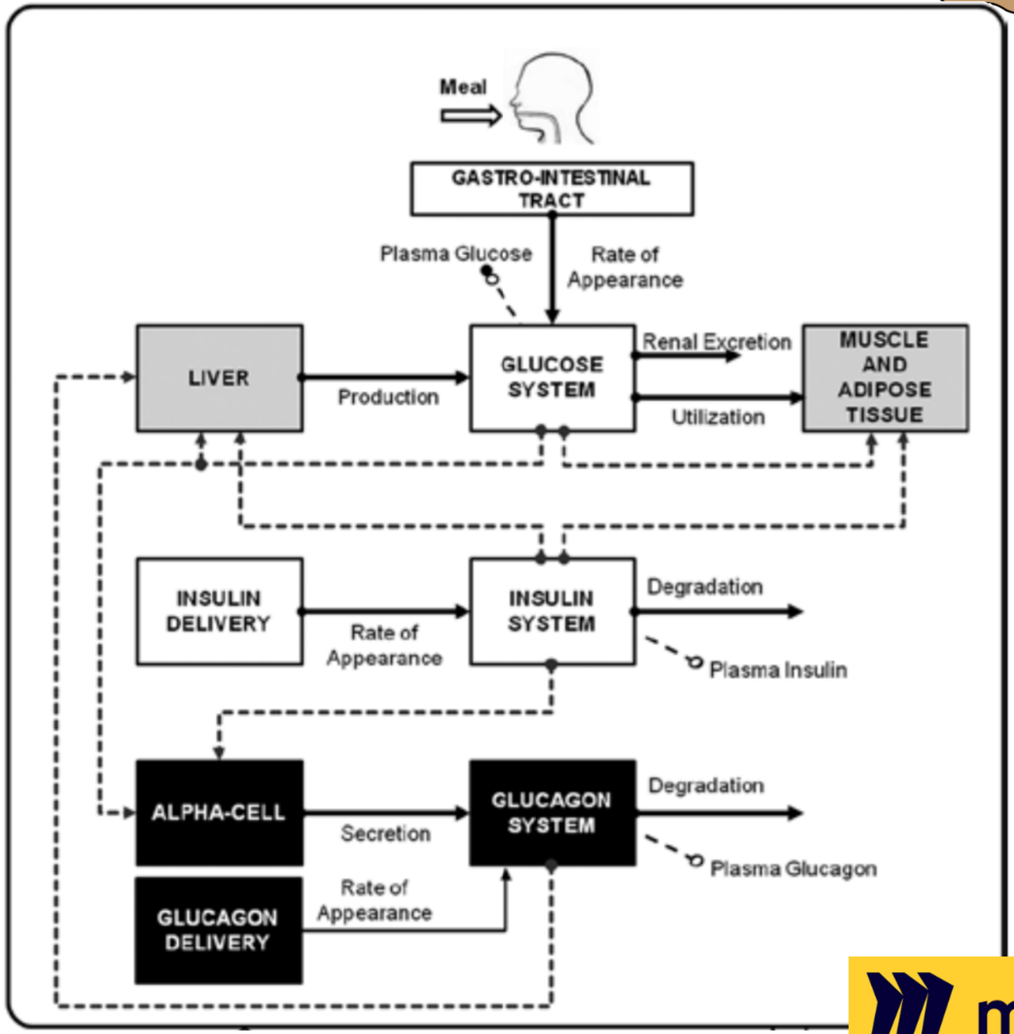
- As system inputs we have the meal glucose equivalent

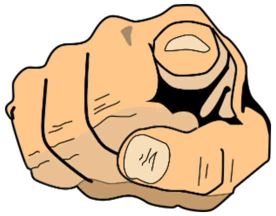
- As system outputs (measurement) we have the glucose measured in tissue (interstitum via CGM)

- To the right a conceptual reference model
  NOTE: not everything will be modelled!
  We are aiminf for a low-order grey box model

- **TASK: Create a Model Specification for the glucose-insulin system. Text & Block Diagramm**

Note: Equations come in SW03





Semester Project – Glucose – Insulin Digital Twin

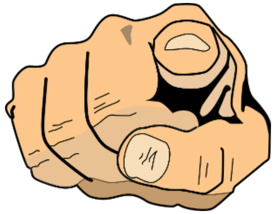
- Analyse the potential value of creating a Digital Twin of the Glucose-Insulin System

- From the point of view of the user: What value could there be in better understanding your Glucose and Insulin levels? What would motivate you to buy a CGM and analyse the Twin’s behaviour.

- Consider that you are a company offering CGM patches & have the opportunity to use Digital Twins. What additional services could you offer using a CGM-Based Digital Twin. What requirements can you derive on your Twin based on these services?

- Use this input to follow the process on slide 4 to derive the 5D Digital Twin requirements.

SW05



Simulation des Blutzucker-Insulin Systems

Essen

Glukose

Körperliche

Aktivität

Nach dem Essen:
Nach 5-10min steigt G
Maximum nach 30-60min

Ohne Essen kehren Blutzucker und Insulin zu den Basalwerten zurück
$G \rightarrow G_b \sim= 5$, $I \rightarrow I_b \sim= 10$
Zeit 2-3 Stunden

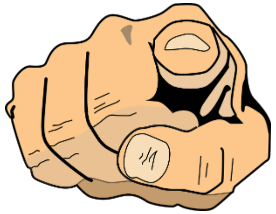
Blutzucker G hoch → Insulin I steigt
Insulin I hoch → Blutzucker G senkt

Typische Werte:
$4 < G < 7$ [mmol/L]
Gefastet: $5 < I < 15$ [μU/mL]
Nach dem Mahlzeit $40 < I < 60$

CGM

Messung

Aufgabe: Basierend auf Ihre System-Modellierung, bauen Sie ein einfaches Model für die Dynamik des Blutzucker-Insulin-Systems (ODEs). Bestätigen Sie das Verhalten mit Simulation.



Insulin Pump

- Consider your use case and the insulin pump in conjunction with your glucose-insulin system model.

- How can you use your glucose-insulin twin in the development of the insulin pump?

- Decide on a use case (only ONE)
  - Additional Digital Twin-based services with the insulin pump
  - Improve development of the insulin pump using Digital Twins

- How does the DT help development
  - What does it replace or enhance?

- What are the benefits & for whom
  - Safety, speed, quality, cost?